

Incident Detection and Analysis Report

Course: Incident Response

Project: SOC Simulation – Wazuh Log Analysis

Author: Matthew Rich

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1. Introduction

This report presents the analysis of Parrot OS logs collected via the Wazuh agent to detect and investigate suspicious activity. The primary focus is on identifying and analyzing a suspicious login attempt, validating Wazuh alerts, and classifying security incidents according to the taught severity matrix. This report also outlines potential security implications of findings.

2. Methodology

The following methodology was used to analyze the logs:

1. Log Collection:

- Logs were collected from the Parrot OS agent (**Parrot0SVM**) via Wazuh.
- Key logs included PAM authentication events (**auth.log**) and **sudo** commands.

2. Event Filtering:

- Logs were filtered to identify unusual sequences, including repeated login session opens/closes and immediate sudo executions.

3. Timeline Creation:

- Events were ordered chronologically to visualize the sequence of actions during the suspicious login attempt.

4. Alert Validation:

- Each log event was cross-referenced with Wazuh rule IDs and alert levels to ensure that alerts triggered correctly on the Wazuh dashboard.

5. Correlation (Planned):

- Ideally, logs from macOS hosts would be correlated by timestamp, username, and IP address to identify coordinated or related events across hosts.

3. Event Timeline – Suspicious Login Attempt

Time (UTC)	Host	Event	Rule ID	Rule Level	Notes
12:20:19.387	ParrotOSV M	Successful sudo to ROOT executed	5402	3	Root access granted immediately after login
12:20:19.393	ParrotOSV M	PAM: Login session opened	5501	3	User session started
12:20:19.399	ParrotOSV M	PAM: Login session closed	5502	3	Session ended quickly – unusual
12:20:37.256	ParrotOSV M	Successful sudo to ROOT executed	5402	3	Repeated root access
12:20:37.258	ParrotOSV M	PAM: Login session opened	5501	3	User session started
12:20:37.260	ParrotOSV M	PAM: Login session closed	5502	3	Quick closure
12:20:45.488	ParrotOSV M	Successful sudo to ROOT executed	5402	3	Pattern continues
12:20:45.496	ParrotOSV M	PAM: Login session opened	5501	3	
12:20:45.510	ParrotOSV M	PAM: Login session closed	5502	3	

Observation: Multiple rapid login opens, sudo executions, and session closures indicate either a misconfigured script or potentially unauthorized activity.

4. Alert Validation

Event	Rule ID	Level	Dashboard Confirmation
PAM: Login session opened	5501	3	Alert triggered on Wazuh dashboard
PAM: Login session closed	5502	3	Alert triggered
Successful sudo to ROOT	5402	3	Alert triggered

All events for the suspicious login attempt generated alerts as expected, validating Wazuh's detection capabilities.

5. Classification of Security Incidents

Incident	Host	Severity	Rule ID	Rationale
Suspicious login sequence	ParrotOSV M	Medium	5501/5502	Rapid login-open-close sessions with immediate sudo execution
Unauthorized sudo execution	ParrotOSV M	High	5402	Root privileges executed immediately after login, posing a privilege escalation risk
Repeated session activity	ParrotOSV M	Low	5501/5502	Multiple short sessions in short time span; may indicate automated activity or testing

6. Security Implications

- **Rapid repeated logins with root access** may indicate a brute-force or automated attack, which could compromise system integrity.
- **Unauthorized sudo executions** present a high risk for privilege escalation if exploited by a malicious actor.

- **Repeated short session activity** could mask malicious activity or indicate misconfiguration that should be corrected.

Recommendation:

- Investigate corresponding macOS logs for related login attempts.
- Verify user accounts and access policies.
- Audit automated scripts or processes to rule out misconfiguration.